Projection Descent for Norm-Attaining Projective Tensors

The gap in the published density lemma and an L-summand repair

Research memorandum

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Abstract

García-Lirola, Guerrero-Viu, and Rueda Zoca asserted that density of projective norm-attaining tensors descends from $X\widehat{\otimes}_\pi Y$ to $X\widehat{\otimes}_\pi Z$ whenever Z is a 1-complemented subspace of Y. The printed proof projects an ambient norm-attaining approximant and treats the projected tensor as norm-attaining. That step is invalid: a contractive projection can destroy projective norm attainment. Moreover, a dense-envelope construction shows that the density statement itself is false in this generality.

This memorandum gives a natural repair. It is enough to assume that Z is an L-summand of Y, equivalently that the projection $P : Y \to Z$ satisfies

$$\|y\| = \|Py\| + \|y - Py\| \qquad (y \in Y).$$

In fact, for $Y = Z \oplus_1 W$, a tensor in

$$X\widehat{\otimes}_\pi Y \cong (X\widehat{\otimes}_\pi Z) \oplus_1 (X\widehat{\otimes}_\pi W)$$

is norm-attaining if and only if both of its components are norm-attaining. Thus the norm-attaining closure decomposes coordinatewise, and density descends exactly.

The repair is close to the natural boundary of standard projection geometry. At the opposite M-summand endpoint, every tensor in $X\widehat{\otimes}_\pi Y$ is the image of a norm-attaining tensor in $X\widehat{\otimes}_\pi(X^* \oplus_\infty Y)$ under the canonical M-projection. Hence even bicontractive M-projections need not preserve norm attainment. The report distinguishes what is proved, what is merely sufficient, and what remains open.

Contents

1 Notation and the printed statement

All Banach spaces are over $\mathbb{K} \in \{\mathbb{R}, \mathbb{C}\}$. A tensor $u \in X \widehat{\otimes}_\pi Y$ is said to attain its projective norm if it admits a representation

$$u = \sum_{n=1}^{\infty} x_n \otimes y_n, \qquad \|u\|_\pi = \sum_{n=1}^{\infty} \|x_n\| \, \|y_n\|.$$

The set of such tensors is denoted by $\mathrm{NA}_\pi(X \widehat{\otimes}_\pi Y)$. The subset of tensors admitting a finite optimal representation is denoted by $\mathrm{FNA}_\pi(X \widehat{\otimes}_\pi Y)$. We write

$$\mathscr{C}_\pi(X, Y) := \overline{\mathrm{NA}_\pi(X \widehat{\otimes}_\pi Y)}^{\|\cdot\|_\pi}.$$

Finite truncation of an optimal representation gives

$$\mathscr{C}_\pi(X, Y) = \overline{\mathrm{FNA}_\pi(X \widehat{\otimes}_\pi Y)}^{\|\cdot\|_\pi}. \tag{1}$$

The published Lemma 4.1 of [4] states that, if

$$\mathscr{C}_\pi(X, Y) = X \widehat{\otimes}_\pi Y$$

and $Z \subseteq Y$ is 1-complemented, then

$$\mathscr{C}_\pi(X, Z) = X \widehat{\otimes}_\pi Z.$$

The exact norm-attainment analogue appearing earlier as Lemma 3.1 in the same paper is correct: if every tensor in $X \widehat{\otimes}_\pi Y$ attains its norm, then every tensor in $X \widehat{\otimes}_\pi Z$ does so. The passage from exact attainment to density is where the argument breaks.

2 The precise gap in the published proof

Let $P : Y \to Z$ be a projection with $\|P\| = 1$, and put

$$Q = \mathrm{Id}_X \otimes P : X \widehat{\otimes}_\pi Y \longrightarrow X \widehat{\otimes}_\pi Z.$$

The canonical inclusion $X \widehat{\otimes}_\pi Z \hookrightarrow X \widehat{\otimes}_\pi Y$ is isometric because Q is a contractive left inverse.

2.1 Why the earlier exact lemma is valid

Suppose $z \in X \widehat{\otimes}_\pi Z$ is norm-attaining when considered in $X \widehat{\otimes}_\pi Y$. Choose an optimal ambient representation

$$z = \sum_{n=1}^{\infty} x_n \otimes y_n, \qquad \sum_n \|x_n\| \, \|y_n\| = \|z\|_{X \widehat{\otimes}_\pi Y}.$$

Since $z = Qz$,

$$z = \sum_{n=1}^{\infty} x_n \otimes P y_n$$

in $X \widehat{\otimes}_\pi Z$. The following chain closes at both ends:

$$\begin{aligned} \|z\|_{X \widehat{\otimes}_\pi Z} &\le \sum_n \|x_n\| \, \|P y_n\| \\ &\le \sum_n \|x_n\| \, \|y_n\| \\ &= \|z\|_{X \widehat{\otimes}_\pi Y} = \|z\|_{X \widehat{\otimes}_\pi Z}. \end{aligned}$$

Hence every inequality is an equality, and the projected representation is optimal. This is the correct content of the published Lemma 3.1.

2.2 The step that fails in Lemma 4.1

For density, the paper starts with $z \in X\widehat{\otimes}_\pi Z$ and chooses an ambient norm-attaining tensor $z' \in X\widehat{\otimes}_\pi Y$ close to z. It then says that, as in Lemma 3.1, Qz' is norm-attaining. But z' need not lie in the range of Q, so $Qz' = z'$ is unavailable.

Indeed, from an optimal representation

$$z' = \sum_n x_n \otimes y_n, \qquad \|z'\|_\pi = \sum_n \|x_n\| \, \|y_n\|,$$

one obtains only

$$\|Qz'\|_\pi \le \sum_n \|x_n\| \, \|Py_n\| \le \sum_n \|x_n\| \, \|y_n\| = \|z'\|_\pi. \tag{2}$$

There is no equality at the right endpoint involving $\|Qz'\|_\pi$, and the first inequality in (2) may be strict. The proof silently assumes precisely the equality that has to be established.

The defect can be written as

$$\delta_P(z'; (x_n, y_n)) := \sum_n \|x_n\| \, \|Py_n\| - \|Qz'\|_\pi \ge 0.$$

Lemma 3.1 forces this defect to vanish because the tensor itself belongs to the projected range. For a general approximant there is no such mechanism.

2.3 Projection can genuinely destroy norm attainment

The missing assertion is not merely unproved; it is false in a particularly strong form.

Theorem 2.1 (Universal M-projection lift)**.** *Let X, Y be Banach spaces and let $z \in X\widehat{\otimes}_\pi Y$. For every representation*

$$z = \sum_{n=1}^{\infty} x_n \otimes y_n, \qquad \sum_n \|x_n\| \, \|y_n\| < \infty,$$

there exists

$$\widetilde{z} \in \mathrm{NA}_\pi\big(X\widehat{\otimes}_\pi(X^* \oplus_\infty Y)\big)$$

such that

$$(\mathrm{Id}_X \otimes R)\widetilde{z} = z,$$

where $R : X^* \oplus_\infty Y \to Y$ is the second-coordinate projection. If the chosen representation of z is finite, then $\widetilde{z}$ belongs to FNA_π.

Proof. Omit zero terms. For each n, choose $x_n^* \in S_{X^*}$ with

$$x_n^*(x_n) = \|x_n\|.$$

Set

$$f_n = \|y_n\| x_n^*, \qquad \widetilde{z} = \sum_n x_n \otimes (f_n, y_n).$$

Since

$$\|(f_n, y_n)\|_{X^* \oplus_\infty Y} = \|y_n\|,$$

the series converges projectively. Define the trace form

$$\tau(x, (f, y)) = f(x).$$

It has norm one, and

$$\tau(x_n, (f_n, y_n)) = \|x_n\| \, \|y_n\|.$$

Therefore
$$\|\widetilde{z}\|_\pi \geq \tau(\widetilde{z}) = \sum_n \|x_n\| \, \|y_n\| \geq \|\widetilde{z}\|_\pi,$$
so the displayed representation is optimal. The projection identity is immediate. Finally,
$$\|(f,y)\|_\infty = \max\{\|(I-R)(f,y)\|, \|R(f,y)\|\},$$
so R is an M-projection, in particular a bicontractive projection. □

Corollary 2.2. *There are a bicontractive M-projection $R : F \to Y$ and a norm-attaining tensor $v \in X\widehat{\otimes}_\pi F$ such that $(\mathrm{Id}_X \otimes R)v$ is not norm-attaining. One may take v to have a finite optimal representation while its image is a finite-rank non-norm-attaining tensor.*

Proof. Non-norm-attaining tensors exist, and finite-rank examples are given in [3, Proposition 5.5]. Apply Theorem 2.1 to such a tensor and to a finite representation in the finite-rank case. □

Thus neither contractivity, bicontractivity, nor the stronger M-projection identity repairs the specific projection step in the printed proof.

3 Why the density statement itself is false

The preceding theorem disproves preservation of individual norm-attaining tensors, but by itself it does not disprove density descent. We record a self-contained dense-envelope construction that does.

3.1 A trace-face correction lemma

Let E, H be Banach spaces and set $G = E^* \oplus_\infty H$. Define
$$\tau_E(e, (e^*, h)) = e^*(e).$$

Lemma 3.1 (Trace-face correction)**.** *If $u \in E\widehat{\otimes}_\pi G$ satisfies*
$$\operatorname{Re}\tau_E(u) = \|u\|_\pi,$$
then $u \in \mathscr{C}_\pi(E, G)$. More precisely, u is approximable by tensors with finite optimal representations supported by τ_E.

Proof. Normalize $\|u\|_\pi = 1$. Fix $k \in \mathbb{N}$ and let $\eta_k > 0$ be a scalar Bishop–Phelps–Bollobás threshold for error $1/k$. Choose a projective representation
$$u = \sum_n \lambda_{k,n} e_{k,n} \otimes (f_{k,n}, h_{k,n})$$
with
$$e_{k,n} \in S_E, \quad (f_{k,n}, h_{k,n}) \in S_G, \quad \sum_n \lambda_{k,n} < 1 + \frac{\eta_k}{k}.$$
The trace equality gives
$$\sum_n \lambda_{k,n}(1 - \operatorname{Re} f_{k,n}(e_{k,n})) < \frac{\eta_k}{k}.$$
Let
$$A_k = \{n : \operatorname{Re} f_{k,n}(e_{k,n}) > 1 - \eta_k\}.$$
Then $\sum_{n \notin A_k} \lambda_{k,n} < 1/k$. For $n \in A_k$, the Bishop–Phelps–Bollobás theorem gives $\widetilde{e}_{k,n} \in S_E$ and $\widetilde{f}_{k,n} \in S_{E^*}$ such that
$$\widetilde{f}_{k,n}(\widetilde{e}_{k,n}) = 1, \quad \|e_{k,n} - \widetilde{e}_{k,n}\| < \frac{1}{k}, \quad \|f_{k,n} - \widetilde{f}_{k,n}\| < \frac{1}{k}.$$

Since $\|h_{k,n}\| \le 1$,

$$\|(\widetilde{f}_{k,n}, h_{k,n})\| = 1.$$

The tensor

$$v_k = \sum_{n \in A_k} \lambda_{k,n} \widetilde{e}_{k,n} \otimes (\widetilde{f}_{k,n}, h_{k,n})$$

is norm-attaining, supported by τ_E, and $\|u - v_k\|_\pi \to 0$. Finite partial sums of v_k are finitely norm-attaining, and a diagonal selection proves the result. □

3.2 Universal graph stabilization

For fixed X, Y, let

$$\Gamma_Y = B_{\mathcal{L}(X,Y^*)}, \qquad \mathcal{U}_X(Y) = Y \oplus_\infty \ell_\infty(\Gamma_Y, X^*).$$

For $B \in \Gamma_Y$, write

$$(B^{\mathsf{t}} y)(x) = B(x)(y).$$

Define

$$j_Y(y) = (y, (B^{\mathsf{t}} y)_{B \in \Gamma_Y}).$$

The map j_Y is an isometry, and its range is 1-complemented by

$$Q_Y(y, (f_B)_B) = j_Y(y).$$

Proposition 3.2 (Universal graph stabilization)**.** *For every $u \in X \widehat{\otimes}_\pi Y$,*

$$(\mathrm{Id}_X \otimes j_Y) u \in \mathscr{C}_\pi(X, \mathcal{U}_X(Y)).$$

Proof. Choose $B_0 \in S_{\mathcal{L}(X,Y^*)}$ with

$$\operatorname{Re} B_0(u) = \|u\|_\pi.$$

The B_0 coordinate of $\mathcal{U}_X(Y)$ is a copy of X^*. Its coordinate trace has norm one and evaluates $(\mathrm{Id}_X \otimes j_Y) u$ as $B_0(u)$. The graph embedding is isometric, so the trace value equals the projective norm. Apply Theorem 3.1. □

Theorem 3.3 (Dense envelope)**.** *For every pair X, Y, there is a Banach space Y_∞ containing Y isometrically as a 1-complemented subspace such that*

$$\mathscr{C}_\pi(X, Y_\infty) = X \widehat{\otimes}_\pi Y_\infty.$$

Proof. Set $Y_0 = Y$ and define inductively

$$Y_{n+1} = \mathcal{U}_X(Y_n),$$

identifying Y_n with its graph image in Y_{n+1}. Every inclusion is isometric and has a contractive retraction. Let Y_∞ be the completion of the increasing union. Compatibility of the retractions makes each Y_n, especially Y_0, 1-complemented in Y_∞.

By Theorem 3.2, every tensor in $X \widehat{\otimes}_\pi Y_n$ belongs to $\mathscr{C}_\pi(X, Y_{n+1})$. Since Y_{n+1} is 1-complemented in Y_∞, finite optimal representations in $X \widehat{\otimes}_\pi Y_{n+1}$ remain optimal in the ambient tensor product. Hence

$$X \widehat{\otimes}_\pi Y_n \subseteq \mathscr{C}_\pi(X, Y_\infty) \qquad (n \in \mathbb{N}).$$

The union of the left-hand spaces is dense in $X \widehat{\otimes}_\pi Y_\infty$, so the closed set $\mathscr{C}_\pi(X, Y_\infty)$ is the whole space. □

Corollary 3.4 (Failure of the printed Lemma 4.1)**.** *There are Banach spaces X, Y, Z such that Y is an isometric 1-complemented subspace of Z,*

$$\mathscr{C}_\pi(X, Z) = X\widehat{\otimes}_\pi Z,$$

but

$$\mathscr{C}_\pi(X, Y) \neq X\widehat{\otimes}_\pi Y.$$

Consequently, Lemma 4.1 of [4] is false as printed.

Proof. Choose a pair X, Y with non-dense norm-attaining tensors, whose existence is proved in [3, Theorem 5.1], and apply Theorem 3.3. Take $Z = Y_\infty$. □

The construction is new and has not undergone independent peer review. The gap in the printed proof, however, is independent of the construction and is already witnessed by Theorem 2.1.

4 The L-summand repair

Definition 4.1. A projection $P : Y \to Y$ is an *L-projection* if

$$\|y\| = \|Py\| + \|y - Py\| \qquad (y \in Y).$$

Its range is called an *L-summand*. Equivalently, if $Z = P(Y)$ and $W = \ker P$, then

$$Y = Z \oplus_1 W$$

isometrically. An M-projection is defined by replacing the sum with the maximum.

The L-identity supplies exactly the equality missing from (2).

Lemma 4.2 (Projective distributivity over an ℓ_1-sum)**.** *For Banach spaces X, Z, W, the canonical map*

$$\Phi : X\widehat{\otimes}_\pi(Z \oplus_1 W) \longrightarrow (X\widehat{\otimes}_\pi Z) \oplus_1 (X\widehat{\otimes}_\pi W)$$

defined on elementary tensors by

$$\Phi(x \otimes (z, w)) = (x \otimes z, x \otimes w)$$

is an onto linear isometry.

Proof. For an elementary tensor,

$$\|x \otimes z\|_\pi + \|x \otimes w\|_\pi = \|x\|(\|z\| + \|w\|) = \|x \otimes (z, w)\|_\pi.$$

The projective definition shows that Φ is contractive. Conversely, the map

$$(u, v) \longmapsto (\mathrm{Id}_X \otimes i_Z)u + (\mathrm{Id}_X \otimes i_W)v$$

is contractive from $(X\widehat{\otimes}_\pi Z) \oplus_1 (X\widehat{\otimes}_\pi W)$ into $X\widehat{\otimes}_\pi(Z \oplus_1 W)$ and is inverse to Φ on a dense algebraic subspace. Hence both maps extend as inverse isometries. □

Theorem 4.3 (Exact decomposition of norm attainment)**.** *Under the isometry in Theorem 4.2, let u correspond to (u_Z, u_W). Then*

$$u \in \mathrm{NA}_\pi(X\widehat{\otimes}_\pi(Z \oplus_1 W)) \quad \Longleftrightarrow \quad u_Z \in \mathrm{NA}_\pi(X\widehat{\otimes}_\pi Z) \text{ and } u_W \in \mathrm{NA}_\pi(X\widehat{\otimes}_\pi W).$$

The same equivalence holds with FNA_π in place of NA_π.

Proof. Assume first that

$$u = \sum_{n=1}^{\infty} x_n \otimes (z_n, w_n)$$

is an optimal representation. Put

$$A = \sum_n \|x_n\| \, \|z_n\|, \qquad C = \sum_n \|x_n\| \, \|w_n\|.$$

Because the norm on $Z \oplus_1 W$ is additive,

$$A + C = \sum_n \|x_n\|(\|z_n\| + \|w_n\|) = \|u\|_\pi.$$

By Theorem 4.2,

$$\|u\|_\pi = \|u_Z\|_\pi + \|u_W\|_\pi.$$

The projected series give

$$\|u_Z\|_\pi \le A, \qquad \|u_W\|_\pi \le C.$$

The sums of the two sides are equal, so both inequalities are equalities. Thus both projected representations are optimal. Finiteness is preserved.

Conversely, concatenate optimal representations of u_Z and u_W, embedding the first in the Z coordinate and the second in the W coordinate. The total coefficient sum is

$$\|u_Z\|_\pi + \|u_W\|_\pi = \|u\|_\pi,$$

so the resulting representation of u is optimal. Again, finite representations remain finite. □

Corollary 4.4 (Closure decomposition)**.** *Under the same identification,*

$$\mathscr{C}_\pi(X, Z \oplus_1 W) = \mathscr{C}_\pi(X, Z) \oplus_1 \mathscr{C}_\pi(X, W).$$

In particular,

$$\mathscr{C}_\pi(X, Z \oplus_1 W) = X \widehat{\otimes}_\pi (Z \oplus_1 W)$$

if and only if

$$\mathscr{C}_\pi(X, Z) = X \widehat{\otimes}_\pi Z \quad \text{and} \quad \mathscr{C}_\pi(X, W) = X \widehat{\otimes}_\pi W.$$

Proof. By Theorem 4.3, the norm-attaining set corresponds exactly to the Cartesian product of the two coordinate norm-attaining sets. Taking closure in an ℓ_1-sum gives the product of the two closures. □

Theorem 4.5 (Corrected density-descent lemma)**.** *Let X, Y be Banach spaces and let Z be an L-summand of Y. If*

$$\mathscr{C}_\pi(X, Y) = X \widehat{\otimes}_\pi Y,$$

then

$$\mathscr{C}_\pi(X, Z) = X \widehat{\otimes}_\pi Z.$$

More strongly, if $P : Y \to Z$ is the L-projection, then

$$(\mathrm{Id}_X \otimes P) \, \mathrm{NA}_\pi(X \widehat{\otimes}_\pi Y) = \mathrm{NA}_\pi(X \widehat{\otimes}_\pi Z),$$

and the same equality holds for FNA_π*.*

Proof. Write $Y = Z \oplus_1 \ker P$ and apply Theorems 4.3 and 4.4. The reverse inclusion in the displayed equality follows because every intrinsic optimal representation remains optimal under the canonical isometric inclusion into the ambient space. □

This is the cleanest direct repair of the printed argument: replace "1-complemented" by "an L-summand."

4.1 Arbitrary ℓ_1-sums

The same proof gives a useful extension. If $(Y_i)_{i\in I}$ is any family, then

$$X\widehat{\otimes}_\pi \left(\bigoplus_{i\in I} Y_i\right)_1 \cong \left(\bigoplus_{i\in I} X\widehat{\otimes}_\pi Y_i\right)_1,$$

and a tensor $(u_i)_i$ is norm-attaining if and only if every nonzero component u_i is norm-attaining. Consequently,

$$\mathscr{C}_\pi\left(X, \left(\bigoplus_{i\in I} Y_i\right)_1\right) = \left(\bigoplus_{i\in I} \mathscr{C}_\pi(X, Y_i)\right)_1.$$

Every element of an $\ell_1(I)$-sum has countable support, so concatenating the coordinate optimal representations causes no set-theoretic difficulty.

4.2 Bidual and integral versions

A Banach space Y is L-embedded if its canonical copy is an L-summand in Y^{**}. Therefore Theorem 4.5 gives the valid bidual replacement

$$\mathscr{C}_\pi(X, Y^{**}) = X\widehat{\otimes}_\pi Y^{**} \quad\Longrightarrow\quad \mathscr{C}_\pi(X, Y) = X\widehat{\otimes}_\pi Y$$

for L-embedded Y.

The 2026 preprint [1] proves that density of classical norm-attaining tensors is equivalent to density of integral norm-attaining tensors. Combining that equivalence with Theorem 4.5 yields the corresponding corrected version of its Corollary 2.14: density of integral norm-attaining tensors descends through L-summands. The statement with an arbitrary 1-complemented subspace is contradicted by Theorem 3.4.

5 How restrictive is the repair?

The phrase "least restrictive" has to be interpreted carefully. There cannot be a unique weakest geometric hypothesis on P: pair-specific assumptions on X, Y, and Z may imply intrinsic density for reasons unrelated to the projection. There is, however, a precise weakest condition for the published proof strategy itself:

$$(\mathrm{Id}_X \otimes P)\,\mathrm{NA}_\pi(X\widehat{\otimes}_\pi Y) \subseteq \mathrm{NA}_\pi(X\widehat{\otimes}_\pi Z).$$

This condition is tautological rather than structural. The L-summand assumption is a standard, checkable geometric condition that forces it, and it does so with an exact coordinate characterization.

The comparison with the two classical absolute-sum endpoints is sharp for the projection mechanism:

- For an ℓ_1-decomposition, every norm-attaining tensor decomposes into norm-attaining coordinates by Theorem 4.3.
- For an ℓ_∞-decomposition, the phenomenon reverses maximally: by Theorem 2.1, the coordinate M-projection maps ambient norm-attaining tensors onto every tensor of the target, including non-norm-attaining ones.

Thus "bicontractively complemented" or "an M-summand" is not enough to validate the sentence used in the published proof. This does not by itself decide whether the density conclusion might hold for all M-summands by some different argument; that narrower question appears to remain open.

For a nonzero factor X, the tensor projection $\mathrm{Id}_X \otimes P$ is an L-projection if and only if P is an L-projection. The nontrivial implication follows by testing the L-identity on $x \otimes y$ with $x \in S_X$:

$$\|y\| = \|x \otimes y\|_\pi = \|x \otimes Py\|_\pi + \|x \otimes (y - Py)\|_\pi = \|Py\| + \|y - Py\|.$$

So the repair cannot be weakened merely by moving the same L-identity from the factor to the tensor product.

6 Conclusions

The status can be summarized precisely.

1. The exact all-tensors descent Lemma 3.1 of [4] is correct for arbitrary 1-complemented subspaces.
2. The density proof for Lemma 4.1 has a definite gap at the claim that the projection of an ambient norm-attaining approximant is norm-attaining.
3. That preservation claim is false even for bicontractive M-projections, by Theorem 2.1.
4. The density statement itself is false for arbitrary 1-complemented subspaces, by the dense-envelope construction.
5. A natural corrected theorem is obtained by requiring the subspace to be an L-summand. Under this assumption, norm attainment and its closure decompose exactly coordinatewise.

The L-summand theorem is therefore a rigorous and near-sharp repair in the standard geometry of contractive projections. It should not be advertised as the logically weakest possible hypothesis, but it is the next natural condition after 1-complementability that restores the missing equality and makes the original proof strategy valid.

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